Final Supply Chain Insights & Recommendations

End-to-End Data Analytics Project Documentation

1. Analytical Overview

An in-depth analysis of 5,000 recent e-commerce shipments was conducted to evaluate the health of the supply chain network. The analysis separated internal fulfillment processing from external carrier transit times to accurately isolate performance bottlenecks.

2. Key Findings & Insights

A. Carrier Reliability Discrepancies

Data indicates a severe reliability gap among shipping partners. **USPS accounts for the highest late delivery rate at ~23%**, significantly dragging down the overall On-Time Delivery (OTD) rate. In contrast, DHL maintained a 95% OTD rate, and FedEx operated at 90%.

B. Internal Warehouse Bottlenecks

While the overall average warehouse processing time is acceptable, the **IL-04 (Midwest) facility shows a processing variance of up to 5 days**. This internal delay guarantees an SLA breach before the package even reaches the carrier.

C. Regional Cost Inefficiencies

Shipments to the 'West' region consistently incur the highest average freight costs ($29.40 avg) combined with the highest transit times. This suggests our current warehouse distribution matrix is poorly optimized for West Coast volume.

3. Strategic Recommendations

- **Carrier Volume Reallocation:** Immediately shift 15% of standard shipment volume from USPS to FedEx or UPS. While USPS base rates are lower, the hidden cost of customer service tickets and refunded shipping related to their 23% late rate negates the upfront savings.
- **Audit Facility IL-04:** Dispatch an operations team to the IL-04 warehouse to conduct a time-in-motion study. The facility requires improved pick-and-pack automation to stabilize processing times below the 2-day threshold.
- **West Coast Fulfillment Expansion:** Consider utilizing a 3PL (Third-Party Logistics) partner in Nevada or California. Fulfilling West Coast orders from TX-03 and IL-04 is driving up freight spend and reducing customer satisfaction.